The phase space of Drell–Yan real emission at fixed rapidity

NLO, NNLO and N^{3}LO in one closed formula

DY_P2B — `phase_space_int/`

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Abstract

The simplest integral of every real-emission family is the one whose integrand is *only* the cut propagators: the phase space of n massless partons plus the colour singlet Q, at fixed invariant mass $Q^2 = m^2$ and fixed rapidity. For the triple real ($n = 3$) it is `INT(topoN,1,1,1,1,1,0,...,0)`, the same in all families. We compute it for any n in $d = 4 - 2\epsilon$, exactly in z, u and ϵ. The result (eq. (11)) is a product of powers and one Gauss hypergeometric function, which equals 1 for $n = 1$. We explain every step, give the ϵ expansion and the soft limit, and compare with the NNLO computation in `nnlo_rr/integrals/IF1/boundaries.nb`. Everything is verified numerically by the scripts in `checks/`.

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Roadmap

The computation has three ideas, one per section:

1. **Factorise** (Sect. 2). The n massless partons only feel the total momentum $K = p_1 + p_2 - Q$ they share. Their phase space is a known function of K^2 alone. What remains is an integral over Q.

2. **Light-cone variables for Q** (Sect. 4). With $Q = a\,p_1 + b\,p_2 + Q_T$ the rapidity constraint is *linear* ($b = u\,a$) and the mass-shell constraint fixes $|Q_T|$. One integral, over a, is left.

3. **Recognise a hypergeometric function** (Sect. 5). The a integrand is a product of three powers. Mapping the integration range to $[0, 1]$ gives Euler's integral for ${}_2F_1$.

The answer is in Sect. 6. The rest of the note covers its use: ϵ expansion, soft limit, the comparison with NNLO, and N^{3}LO.

1 The integral and what each factor means

1.1 Definition

Incoming massless partons p_1, p_2, with $s = (p_1 + p_2)^2 = 2\,p_1 \cdot p_2$. Outgoing massless partons $k_1, \ldots, k_n$. The colour singlet carries what is left:

$$Q = p_1 + p_2 - \sum_{i=1}^{n} k_i .$$

We study

$$\boxed{\Phi_n(z,u) = \int \prod_{i=1}^{n} \mathrm{d}^d k_i\, \delta^+(k_i^2)\;\; \delta^+(Q^2 - m^2)\;\; \delta(Q\cdot p_1 - u\,Q\cdot p_2)} \qquad z = \frac{m^2}{s}, \qquad d = 4 - 2\epsilon. \tag{1}$$

Here $\delta^+(k^2) = \delta(k^2)\,\theta(k^0)$. Each factor has a meaning:

- $\delta^+(k_i^2)$: parton i is real (on shell, positive energy). These are the cut propagators $1/k_i^2$.
- $\delta^+(Q^2 - m^2)$: the singlet has invariant mass m, i.e. we look at fixed Q^2. This is the cut of $1/(Q^2 - m^2)$.
- $\delta(Q\cdot p_1 - u\,Q\cdot p_2)$: the singlet has fixed rapidity (see below). This is the fifth "propagator" of our families. It is linear in the momenta.

No factors of 2π, i or $e^{\epsilon\gamma_E}$ are included. Any other normalisation is an overall ϵ-dependent constant; Sect. 9 identifies the one used at NNLO.

1.2 Why the last delta fixes the rapidity

Work in the partonic centre-of-mass frame, with $p_1 = \frac{\sqrt{s}}{2}(1,0,0,1)$ and $p_2 = \frac{\sqrt{s}}{2}(1,0,0,-1)$. Write $Q = (m_T \cosh Y, \vec{Q}_T, m_T \sinh Y)$ with $m_T^2 = m^2 + Q_T^2$. Then

$$Q \cdot p_1 = \frac{\sqrt{s}}{2}\,(Q^0 - Q^3) = \frac{\sqrt{s}}{2}\,m_T\,e^{-Y}, \qquad Q \cdot p_2 = \frac{\sqrt{s}}{2}\,(Q^0 + Q^3) = \frac{\sqrt{s}}{2}\,m_T\,e^{Y},$$

so $Q \cdot p_1 - u\,Q \cdot p_2 = 0$ means $u = e^{-2Y}$: *u is the rapidity of Q*. Note that m_T drops out of the ratio.

The "different delta". One can also fix the rapidity with $\delta(u - u_Q)$, $u_Q = Q \cdot p_1 / Q \cdot p_2$. The two are related by $\delta(Q \cdot p_1 - u\,Q \cdot p_2) = \delta(u - u_Q)/(Q \cdot p_2)$, since the derivative of the argument with respect to u is $Q \cdot p_2$. So (1) is the u-distribution of the phase space weighted with $1/Q \cdot p_2$. This is the factor `1/p2dotq` of the NNLO notebook.

1.3 Physical region and dimension

The rapidity of an object of mass m produced at energy $\sqrt{s}$ is bounded by $|Y| \le \frac{1}{2}\ln\frac{1}{z}$. In terms of u:

$$z \le u \le 1/z.$$

We will re-derive this below, as the condition for the a integral to be non-empty.

Dimension check: each $\int \mathrm{d}^d k\,\delta^+(k^2)$ has mass dimension $d-2$, and each of the two other deltas has -2. So $[\Phi_n] = n(d-2) - 4 = 2(n - 2 - n\epsilon)$, and $\Phi_n \propto s^{\,n-2-n\epsilon}$ times a function of z and u.

1.4 Link with our integral families

In `Code_RRR` every momentum is incoming: $r_i = -k_i$, $q = -Q$. The five cut propagators are

$$r_1^2, \quad r_2^2, \quad r_3^2, \quad (p_1+p_2+r_1+r_2+r_3)^2 - m^2 = Q^2 - m^2, \quad -q\cdot p_1 + u\,q\cdot p_2 = Q\cdot p_1 - u\,Q\cdot p_2.$$

Hence the sector with only the cut lines,

$$\texttt{INT(topoN,1,1,1,1,1,0,0,0,0,0,0,0)},$$

is Φ_3 for *every* family, up to the normalisation convention. The pipeline uses $m^2 = 1$, so $s = 1/z$.

2 Strategy: factorise off the massless partons

Insert $1 = \int \mathrm{d}^d Q\,\delta^d\big(Q - p_1 - p_2 + \sum_i k_i\big)$ into (1). The partons now only appear in

$$\Phi_n^{(0)}(K^2) \equiv \int \prod_{i=1}^{n} \mathrm{d}^d k_i\,\delta^+(k_i^2)\;\delta^d\Big(K - \sum_i k_i\Big), \qquad K = p_1 + p_2 - Q, \tag{2}$$

the phase space of n massless particles with total momentum K. It is Lorentz invariant, so it depends only on K^2. Therefore

$$\Phi_n = \int \mathrm{d}^d Q\;\delta^+(Q^2 - m^2)\;\delta(Q\cdot p_1 - u\,Q\cdot p_2)\;\Phi_n^{(0)}\big((p_1+p_2-Q)^2\big). \tag{3}$$

Two tasks remain: compute $\Phi_n^{(0)}(K^2)$ (Sect. 3) and the Q integral (Sect. 4).

3 The massless n-body phase space

We need the solid angle of the unit sphere in D dimensions, $\Omega_D = 2\pi^{D/2}/\Gamma(D/2)$ (e.g. $\Omega_3 = 4\pi$).

3.1 Two bodies, one of them massive

Take one massless particle k and one of mass M, in the rest frame of K ($K = (\sqrt{K^2}, \vec{0})$):

$$\Phi_2(K^2; M^2, 0) = \int \mathrm{d}^d k\, \delta^+(k^2)\, \delta^+\big((K-k)^2 - M^2\big) = \int \frac{\mathrm{d}^{d-1}k}{2|\vec{k}|}\, \delta\big(K^2 - 2\sqrt{K^2}\,|\vec{k}| - M^2\big).$$

In the first step $\int \mathrm{d}k^0\, \delta^+(k^2) = 1/(2|\vec{k}|)$, and we used $(K-k)^2 = K^2 - 2\sqrt{K^2}|\vec{k}|$ for massless k. In spherical coordinates, $\mathrm{d}^{d-1}k = \Omega_{d-1}|\vec{k}|^{d-2}\mathrm{d}|\vec{k}|$. The delta fixes $|\vec{k}| = E \equiv (K^2 - M^2)/(2\sqrt{K^2})$, with Jacobian $1/(2\sqrt{K^2})$. This gives

$$\Phi_2(K^2; M^2, 0) = \frac{\Omega_{d-1}}{4\sqrt{K^2}}\, E^{\,d-3} = \frac{\pi^{1-\epsilon}\,\Gamma(1-\epsilon)}{2\,\Gamma(2-2\epsilon)}\, (K^2)^{-1+\epsilon}\, (K^2 - M^2)^{1-2\epsilon},$$

where the duplication formula $\Gamma(1-\epsilon)\Gamma(\frac{3}{2}-\epsilon) = 2^{-1+2\epsilon}\sqrt{\pi}\,\Gamma(2-2\epsilon)$ was used to write $\Omega_{3-2\epsilon} = 2^{2-2\epsilon}\pi^{1-\epsilon}\Gamma(1-\epsilon)/\Gamma(2-2\epsilon)$. Two massless particles: set $M = 0$, $\Phi_2^{(0)}(K^2) = \frac{\pi^{1-\epsilon}\Gamma(1-\epsilon)}{2\Gamma(2-2\epsilon)}(K^2)^{-\epsilon}$.

3.2 Recursion to n bodies

Group the first $n-1$ partons into one "particle" of mass M: insert $1 = \int \mathrm{d}M^2 \int \mathrm{d}^d P\, \delta^+(P^2 - M^2)\, \delta^d(P - \sum_{i<n} k_i)$. Then

$$\Phi_n^{(0)}(K^2) = \int_0^{K^2} \mathrm{d}M^2\; \Phi_2(K^2; M^2, 0)\; \Phi_{n-1}^{(0)}(M^2).$$

Assume $\Phi_{n-1}^{(0)}(M^2) = c_{n-1}(M^2)^{n-3-(n-2)\epsilon}$. With $M^2 = K^2 t$ the integral is a Beta function, $\int_0^1 \mathrm{d}t\, t^{\,n-3-(n-2)\epsilon}(1-t)^{1-2\epsilon} = B\big((n-2)(1-\epsilon), 2-2\epsilon\big)$, and

$$\Phi_n^{(0)}(K^2) = c_n\, (K^2)^{\,n-2-(n-1)\epsilon}, \qquad c_n = \frac{\pi^{(n-1)(1-\epsilon)}\,\Gamma(1-\epsilon)^n}{2^{n-1}\,\Gamma\big((n-1)(1-\epsilon)\big)\,\Gamma\big(n(1-\epsilon)\big)} \quad (n \geq 2). \tag{4}$$

The cases needed:

$$\Phi_1^{(0)} = \delta^+(K^2), \qquad c_2 = \frac{\pi^{1-\epsilon}\Gamma(1-\epsilon)}{2\Gamma(2-2\epsilon)}, \qquad c_3 = \frac{\pi^{2-2\epsilon}\Gamma(1-\epsilon)^3}{4\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)}.$$

The power of K^2 follows from dimensions: $n(d-2) - d = 2(n - 2 - (n-1)\epsilon)$.

4 The integral over Q in light-cone variables

4.1 Why light-cone variables

Both remaining constraints are simple in the basis $\{p_1, p_2, \text{transverse}\}$:

$$Q = a\, p_1 + b\, p_2 + Q_T, \qquad Q_T \cdot p_1 = Q_T \cdot p_2 = 0, \qquad Q_T^2 = -q_T^2 \;\; (q_T \geq 0). \tag{5}$$

Since $p_1^2 = p_2^2 = 0$:

$$Q \cdot p_1 = \tfrac{s}{2}\, b, \qquad Q \cdot p_2 = \tfrac{s}{2}\, a, \qquad Q^2 = s\, a\, b - q_T^2.$$

The rapidity constraint is linear in (a, b) and the mass-shell constraint involves q_T only through q_T^2.

4.2 The measure

In the frame of Sect. 1, $Q^0 = \frac{\sqrt{s}}{2}(a+b)$ and $Q^3 = \frac{\sqrt{s}}{2}(a-b)$. Hence

$$\mathrm{d}Q^0\mathrm{d}Q^3 = \frac{s}{4}\left|\det\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\right|\mathrm{d}a\,\mathrm{d}b = \frac{s}{2}\,\mathrm{d}a\,\mathrm{d}b,$$

and

$$\mathrm{d}^dQ = \frac{s}{2}\,\mathrm{d}a\,\mathrm{d}b\,\mathrm{d}^{d-2}q_T. \tag{6}$$

4.3 Doing the two deltas

Rapidity. $\delta(Q\cdot p_1 - u\,Q\cdot p_2) = \delta\big(\frac{s}{2}(b-u\,a)\big) = \frac{2}{s}\,\delta(b-u\,a)$. The b integral sets $b = u\,a$, and the $\frac{2}{s}$ cancels the $\frac{s}{2}$ of the measure.

Mass shell. Now $Q^2 = s\,u\,a^2 - q_T^2$. In the $(d-2)$-dimensional Euclidean transverse space, $\mathrm{d}^{d-2}q_T = \frac{\Omega_{d-2}}{2}(q_T^2)^{\frac{d-4}{2}}\mathrm{d}q_T^2$, so

$$\int \mathrm{d}^{d-2}q_T\;\delta\big(sua^2 - q_T^2 - m^2\big) = \frac{\pi^{1-\epsilon}}{\Gamma(1-\epsilon)}\left(sua^2-m^2\right)^{-\epsilon}, \tag{7}$$

with $q_T^2 = sua^2 - m^2$ fixed. At $\epsilon = 0$ this is just π: in 4 dimensions the transverse "sphere" is a circle and its length times the Jacobian is constant. The factor $(\dots)^{-\epsilon}$ is the d-dimensional remnant.

The recoil. With $b = ua$ and $q_T^2 = sua^2 - m^2$, the momentum of the partons $K = (1-a)p_1 + (1-ua)p_2 - Q_T$ has

$$K^2 = s(1-a)(1-ua) - q_T^2 = s\big(1-a(1+u)\big) + m^2 = s(1+u)\,(a_1 - a), \qquad a_1 \equiv \frac{1+z}{1+u}.$$

4.4 Integration limits and the physical region

a is bounded on both sides:

- $q_T^2 \geq 0$ requires $a \geq a_0 \equiv \sqrt{z/u}$; $a > 0$ comes from $Q^0 > 0$;
- $K^2 \geq 0$ requires $a \leq a_1$; then also $K^0 = \frac{\sqrt{s}}{2}(2 - a(1+u)) > 0$, since $a_1(1+u) = 1+z < 2$.

The range is non-empty iff $a_0 \leq a_1$. Multiplying by $(1+u)$, this is $w \geq 0$, where

$$w \equiv (1+u)(a_1 - a_0) = 1 + z - (1+u)\sqrt{z/u} = \left(1 - \sqrt{\tfrac{z}{u}}\right)\left(1 - \sqrt{zu}\right). \tag{8}$$

Both factors are ≥ 0 exactly for $z \leq u \leq 1/z$ (both negative is impossible for $z < 1$). This re-derives the physical region; $w \to 0$ at its edges, the maximal $|Y|$.

4.5 Result: a one-dimensional integral

Collect everything. Use $sua^2 - m^2 = su\,(a-a_0)(a+a_0)$ and $\Phi_n^{(0)}(K^2) = c_n[s(1+u)(a_1-a)]^k$ with $k \equiv n - 2 - (n-1)\epsilon$:

$$\Phi_n = \frac{\pi^{1-\epsilon}}{\Gamma(1-\epsilon)}\,c_n\,s^{\,n-2-n\epsilon}\,u^{-\epsilon}(1+u)^k\int_{a_0}^{a_1}\mathrm{d}a\,(a-a_0)^{-\epsilon}\,(a+a_0)^{-\epsilon}\,(a_1-a)^k. \tag{9}$$

For $n = 1$, $\Phi_1^{(0)} = \delta(K^2)$ instead puts $a = a_1$ and no integral is left (Sect. 6).

5 The hypergeometric function

Map the range to $[0,1]$ with $a = a_0 + (a_1 - a_0)\,t$:

$$a - a_0 = (a_1 - a_0)\,t, \qquad a_1 - a = (a_1 - a_0)(1-t), \qquad a + a_0 = 2a_0\,(1 + x\,t), \qquad x \equiv \frac{a_1 - a_0}{2a_0}.$$

The integral becomes

$$(a_1 - a_0)^{\,n-1-n\epsilon}\,(2a_0)^{-\epsilon} \int_0^1 \mathrm{d}t\; t^{-\epsilon}(1-t)^k (1 + x\,t)^{-\epsilon}.$$

The two powers of t and $1-t$ alone would give a Beta function. The extra $(1+xt)^{-\epsilon}$ turns it into Euler's integral representation

$$\int_0^1 \mathrm{d}t\, t^{\beta-1}(1-t)^{\gamma-\beta-1}(1-wt)^{-\alpha} = B(\beta, \gamma-\beta)\; {}_2F_1(\alpha, \beta; \gamma; w),$$

here with $\alpha = \epsilon$, $\beta = 1-\epsilon$, $\gamma = n - n\epsilon$, $w = -x$:

$$\int_{a_0}^{a_1} \mathrm{d}a \cdots = (a_1 - a_0)^{\,n-1-n\epsilon}\,(2a_0)^{-\epsilon}\; B\big(1-\epsilon, (n-1)(1-\epsilon)\big)\; {}_2F_1\big(\epsilon, 1-\epsilon; n - n\epsilon; -x\big). \tag{10}$$

The variable x measures how far the integration range extends compared with its distance from the singular point $a = -a_0$. Near threshold or at the edges of the rapidity range $x \to 0$ and ${}_2F_1 \to 1$.

6 The result

Two identities make the result compact. First, $u(a_1 - a_0)(a_1 + a_0) = u\,a_1^2 - z = \frac{(u-z)(1-uz)}{(1+u)^2}$ (multiply out). Second, $(2a_0)^{-\epsilon} = (a_1 + a_0)^{-\epsilon}(1+x)^{\epsilon}$. Using them, together with $(1+u)^k (a_1 - a_0)^{(n-1)(1-\epsilon)} = w^{(n-1)(1-\epsilon)}/(1+u)$:

$$\boxed{\Phi_n(z,u) = C_n\; s^{\,n-2-n\epsilon}\; \frac{w^{(n-1)(1-\epsilon)}}{1+u}\; \Big[\frac{(u-z)(1-uz)}{(1+u)^2}\Big]^{-\epsilon}\; G_n(\epsilon, x)} \tag{11}$$

$$G_n(\epsilon, x) = (1+x)^{\epsilon}\; {}_2F_1\big(\epsilon, 1-\epsilon; n - n\epsilon; -x\big), \qquad C_n = \frac{\pi^{n(1-\epsilon)}\,\Gamma(1-\epsilon)^n}{2^{n-1}\,\Gamma\big(n(1-\epsilon)\big)^2},$$

$$w = \Big(1 - \sqrt{\tfrac{z}{u}}\Big)\Big(1 - \sqrt{zu}\Big), \qquad x = \frac{w\sqrt{u}}{2\sqrt{z}\,(1+u)}. \tag{12}$$

Here $C_n = \frac{\pi^{1-\epsilon}}{\Gamma(1-\epsilon)}\, c_n\, B(1-\epsilon, (n-1)(1-\epsilon))$. The structure of (11):

- $w^{(n-1)(1-\epsilon)}$: the "available phase space" for the partons. It vanishes at the kinematic boundary; each extra parton adds one power.
- $[(u-z)(1-uz)/(1+u)^2]^{-\epsilon}$: from the transverse momentum of Q in d dimensions. At NLO this is the well-known $[(u-z)(1-uz)]^{-\epsilon}$.
- G_n: the only non-trivial function, $G_n = 1 + O(\epsilon\, x)$.

6.1 The three orders

- $n = 1$ **(NLO).** Here $\delta(K^2) = \delta\big(s(1+u)(a_1 - a)\big)$ sets $a = a_1$. Directly from (7):

$$\Phi_1 = \frac{\pi^{1-\epsilon}}{\Gamma(1-\epsilon)}\,\frac{s^{-1-\epsilon}}{1+u}\,\Big[\frac{(u-z)(1-uz)}{(1+u)^2}\Big]^{-\epsilon}.$$

  This is (11) with $C_1 = \pi^{1-\epsilon}/\Gamma(1-\epsilon)$, $w^0 = 1$ and $G_1 = 1$. Consistently, ${}_2F_1(\epsilon, 1-\epsilon; 1-\epsilon; -x) = (1+x)^{-\epsilon}$, so $G_1 = 1$ also from the formula. At NLO the deltas fix everything; nothing is left to integrate.

- $n = 2$ **(NNLO).** $\Phi_2 = \frac{\pi^{2-2\epsilon}\Gamma(1-\epsilon)^2}{2\Gamma(2-2\epsilon)^2}\,s^{-2\epsilon}\,\frac{w^{1-\epsilon}}{1+u}\,\big[\frac{(u-z)(1-uz)}{(1+u)^2}\big]^{-\epsilon} G_2(\epsilon, x)$.

- $n = 3$ **(N^{3}LO).**

$$\Phi_3 = \frac{\pi^{3-3\epsilon}\Gamma(1-\epsilon)^3}{4\,\Gamma(3-3\epsilon)^2}\,s^{1-3\epsilon}\,\frac{w^{2-2\epsilon}}{1+u}\Big[\frac{(u-z)(1-uz)}{(1+u)^2}\Big]^{-\epsilon}(1+x)^{\epsilon}\,{}_2F_1(\epsilon, 1-\epsilon; 3-3\epsilon; -x).$$

6.2 Sanity checks you can do by eye

- $\epsilon = 0$**:** $G_n = 1$ and $\Phi_n = C_n\, s^{n-2} w^{n-1}/(1+u)$, e.g. $\Phi_3 = \frac{\pi^3}{16}\, s\, \frac{w^2}{1+u}$.
- **Dimension:** $s^{\,n-2-n\epsilon}$, as counted in Sect. 1.
- **Edges of the rapidity range:** $w \to 0$ at $u \to z$ and $u \to 1/z$; the phase space closes.
- $p_1 \leftrightarrow p_2$**:** exchanging the beams sends $Y \to -Y$, i.e. $u \to 1/u$. From the delta, $\delta(Q\cdot p_1 - \frac{1}{u} Q\cdot p_2) = u\,\delta(Q\cdot p_2 - u\,Q\cdot p_1)$, so one needs $\Phi_n(z, 1/u) = u\,\Phi_n(z,u)$. In (11), w, x and $(u-z)(1-uz)/(1+u)^2$ are invariant under $u \to 1/u$, and $\frac{1}{1+1/u} = \frac{u}{1+u}$. ✓

7 The ϵ expansion

All factors of (11) except G_n are powers: they expand into logarithms of w, $1+u$, $(u-z)(1-uz)$ and constants. For G_n, expand the series of ${}_2F_1$ or the Euler integral. With $L = \ln(1+x)$:

$$G_2 = 1 + \epsilon\Big(1 - \frac{L}{x}\Big) + \epsilon^2\,\frac{6x - 2L + xL^2 + 2(2+x)\mathrm{Li}_2(-x)}{2x} + O(\epsilon^3), \tag{13}$$

$$G_3 = 1 + \epsilon\,\frac{x(2+3x) - 2(1+2x)L}{2x^2} + \epsilon^2\,\frac{x(7+11x) - (1+2x)L + (1+2x+2x^2)L^2 + 2(3+6x+2x^2)\mathrm{Li}_2(-x)}{2x^2} + O(\epsilon^3). \tag{14}$$

How the $O(\epsilon)$ term arises: $(\epsilon)_m \approx \epsilon\,(m-1)!$ for small ϵ, so ${}_2F_1(\epsilon, 1-\epsilon; n; -x) \approx 1 + \epsilon \sum_{m\geq 1} \frac{(m-1)!}{(n)_m}(-x)^m$. For $n = 2$ the sum is $\sum_m \frac{(-x)^m}{m(m+1)} = 1 - \frac{(1+x)L}{x}$; adding ϵL from $(1+x)^{\epsilon}$ gives $1 - L/x$. These coefficients are regular at $x = 0$ (e.g. $1 - L/x = x/2 + O(x^2)$), so $G_n - 1$ is small near threshold. Higher orders: expand the Euler integral; this gives harmonic polylogarithms of x (HypExp or PolyLogTools).

8 NNLO variables and the soft limit

8.1 The variables of the NNLO computation

The NNLO computation uses $\bar{z} = 1 - z$ and $y \in [0,1]$, with

$$u = \frac{y + z - yz}{1 - y + yz} \qquad (y = 0 \Rightarrow u = z, \quad y = 1 \Rightarrow u = 1/z).$$

So y maps the rapidity range onto $[0,1]$. A short computation gives the exact relations

$$1+u=\frac{1+z}{1-y\bar{z}},\qquad u-z=\frac{y(1-z^2)}{1-y\bar{z}},\qquad 1-uz=\frac{(1-y)(1-z^2)}{1-y\bar{z}},$$
$$\frac{(u-z)(1-uz)}{(1+u)^2}=y(1-y)\,\bar{z}^{\,2}. \tag{15}$$

So the bracket in (11) is rational in (y,z); the square roots only sit in w and x. These are the roots the NNLO notebook handles with $r=\sqrt{u}$, $t=\sqrt{z}$.

8.2 Soft limit

$\bar{z}\to 0$ is the threshold: no energy is left for radiation. At fixed y one finds $\sqrt{z/u}\simeq 1-y\bar{z}$ and $\sqrt{zu}\simeq 1-(1-y)\bar{z}$, hence

$$w=y(1-y)\bar{z}^2\big(1+\tfrac{\bar{z}}{2}+O(\bar{z}^2)\big),\qquad x=\tfrac{1}{4}\,y(1-y)\bar{z}^2+O(\bar{z}^3),\qquad G_n=1+O(\epsilon\,\bar{z}^2),\qquad 1+u\to 2.$$

With $m^2=1$, $s^{\cdots}=z^{-\cdots}\to 1$, and the leading power is

$$\Phi_n^{\mathrm{LP}}=\frac{C_n}{2}\,\big(y(1-y)\big)^{\,n-1-n\epsilon}\,\bar{z}^{\,2(n-1)-2n\epsilon}. \tag{16}$$

The exponents count the phase space: each parton contributes $\bar{z}^{2-2\epsilon}$ (energy and angles scale with $\bar{z}$), and $\big(y(1-y)\bar{z}^2\big)^{-\epsilon}$ comes from the transverse momentum of Q.

9 Comparison with the NNLO computation

File: `nnlo_rr/integrals/IF1/boundaries.nb`, section "phase space". There the phase-space master $j[\mathrm{IF1},1,1,1,1,1,0,0,0]$ is integrated over angular variables $\lambda_1,\lambda_2,\lambda_4$ and expanded at leading power in $\bar{z}$. The exported result is the boundary condition of the canonical differential equations.

9.1 (a) The exported soft master is right

The file `SoftMasters/m1_1111000.m` contains `P[bz,2-4eps] P2[(1-y)y,1-2eps]` times a series in ϵ up to ϵ^6. The prefactor is exactly the structure of (16) for $n=2$. Its coefficients agree *exactly, through* ϵ^6 with

$$\frac{e^{2\epsilon\gamma_E}\,\Gamma(1-\epsilon)^2}{4\,\Gamma(2-2\epsilon)^2}=\frac{e^{2\epsilon\gamma_E}}{\pi^{2-2\epsilon}}\times\frac{C_2}{2}=\frac{1}{4}+\epsilon+\epsilon^2\Big(3-\frac{\pi^2}{8}\Big)+\dots \tag{17}$$

So the NNLO boundary is the leading power of the exact Φ_2. This also fixes the normalisation used at NNLO: $\mathcal{N}_2=e^{2\epsilon\gamma_E}/\pi^{2-2\epsilon}$ times (1), i.e. a factor $e^{\epsilon\gamma_E}/\pi^{1-\epsilon}$ per real parton.

9.2 (b) The closed formula in the comment is exact only at leading power

The notebook also contains, in a comment ("result to print"), a closed formula

$$F_c=\frac{\big((u-z)(1-uz)\big)^{1-2\epsilon}}{(1+u)^{1-2\epsilon}\,\big(\sqrt{u}(\sqrt{u}+\sqrt{z})(1+\sqrt{uz})\big)^{1-\epsilon}}\;\times(\text{constants})\times K[0].$$

To compare it with (11), note $(u-z)=(\sqrt{u}-\sqrt{z})(\sqrt{u}+\sqrt{z})$ and $(1-uz)=(1-\sqrt{uz})(1+\sqrt{uz})$, so

$$\frac{(u-z)(1-uz)}{\sqrt{u}(\sqrt{u}+\sqrt{z})(1+\sqrt{uz})}=\frac{(\sqrt{u}-\sqrt{z})(1-\sqrt{uz})}{\sqrt{u}}=w.$$

Hence $F_c = \frac{w^{1-\epsilon}}{1+u}\left[\frac{(u-z)(1-uz)}{(1+u)^2}\right]^{-\epsilon}$: *exactly* (11) *for* $n=2$ *without the factor* G_2:

$$\frac{\Phi_2}{C_2\, s^{-2\epsilon} F_c} = G_2(\epsilon, x) = 1 + \epsilon\Big(1 - \frac{\ln(1+x)}{x}\Big) + O(\epsilon^2) = 1 + \frac{\epsilon\, x}{2} + \ldots \;\simeq\; 1 + \frac{\epsilon}{8}\, y(1-y)\, \bar{z}^2. \quad (18)$$

Therefore F_c is

- exact at $\epsilon = 0$, for all z;
- exact at leading power in $\bar{z}$, to all orders in ϵ. This is all that was exported and used, so the NNLO results are not affected;
- not exact for general z beyond $O(\epsilon^0)$: it misses the relative correction $G_2 - 1 \simeq \epsilon x/2$, a power-suppressed $O(\epsilon\, \bar{z}^2)$ effect.

Numerically, at $\epsilon = 0.3$ (the middle column is a direct numerical integration of (9)):

$\bar{z}$	y	x	$\Phi_2/(C_2 s^{-2\epsilon} F_c)$ (numerical)	$G_2(\epsilon, x)$	$1 + \epsilon x/2$
1/2	1/5	$1.96 \cdot 10^{-2}$	1.0029115462	1.0029115462	1.00294
1/2	1/2	$3.03 \cdot 10^{-2}$	1.0044765289	1.0044765289	1.00455
1/10	1/5	$4.44 \cdot 10^{-4}$	1.0000666211	1.0000666211	1.000067
1/10	1/2	$6.94 \cdot 10^{-4}$	1.0001040555	1.0001040555	1.000104
1/100	1/5	$4.04 \cdot 10^{-6}$	1.0000006061	1.0000006061	1.00000061
1/100	1/2	$6.31 \cdot 10^{-6}$	1.0000009470	1.0000009470	1.00000095

The deviation falls like $\bar{z}^2$ and is already 10^{-4} at $\bar{z} = 0.1$. This is why it is invisible in the soft expansion but would matter if F_c were used as the full master.

9.3 (c) Is it a matter of which delta?

A natural suspicion: maybe F_c is not the master but the phase space with the *other* rapidity delta, $\delta(u - u_Q)$, which differs by the factor $Q \cdot p_2$ inside the integral (Sect. 1). We tested this directly, integrating the notebook's own `INTEGRAND` (copied verbatim, with the (y, z)-dependent part of `Norm`) over $\lambda_1, \lambda_2, \lambda_4$. The λ_4 and λ_2 integrals factorise into Beta functions, and only λ_1 is left, which we integrate numerically. Two objects:

$$A = \int \texttt{INTEGRAND} \quad (\text{delta } \delta(u - u_Q)), \qquad B = \int \texttt{INTEGRAND}/Q{\cdot}p_2 \quad (\text{the master}).$$

Ratios over five points (z, u); a ratio that is the same at every point means the two functions coincide up to an ϵ-dependent constant:

ratio	$\epsilon = 0$	$\epsilon = 0.3$
A / (9) with extra factor $Q \cdot p_2 = \frac{as}{2}$	2π everywhere	36.0874… everywhere
B / (9) (the master)	2π everywhere	36.0874… everywhere
A / F_c	1.34 … 2.91	14.7 … 31.7
B / F_c	2π everywhere	68.5 … 72.7

Conclusions:

- The notebook's parametrisation is *exact for all* z, for both deltas: A and B agree with the light-cone results everywhere, not only in the soft limit. The factor $1/Q \cdot p_2$ pulled out of the delta is handled correctly.
- F_c is not the $\delta(u - u_Q)$ phase space: A/F_c varies already at $\epsilon = 0$.

- F_c is the master at $\epsilon = 0$ (and at leading power), as found in (b).
- The most likely reading is that `K[0]` in the comment stands for the leftover λ_1 integral. Then the formula is exact, with `K[0]` $\propto B(1-\epsilon, 1-\epsilon)\, G_2(\epsilon, x)$: a function of z, u that becomes a constant only at leading power in $\bar{z}$. That leading power is what the notebook evaluates (`Series[...,{bz,0,0}]`) and exports.

10 N^{3}LO: what to use

- **The master.** `INT(topoN,1,1,1,1,1,0,...,0)` $= \Phi_3$ of (11), exact. In the normalisation analogous to NNLO, $\mathcal{N}_3 = e^{3\epsilon\gamma_E}/\pi^{3-3\epsilon}$ (an assumption, to be matched to the convention of the reduction and the other masters):

$$\mathcal{N}_3\,\Phi_3 = \frac{e^{3\epsilon\gamma_E}\Gamma(1-\epsilon)^3}{4\,\Gamma(3-3\epsilon)^2}\; s^{1-3\epsilon}\;\frac{w^{2-2\epsilon}}{1+u}\left[y(1-y)\bar{z}^2\right]^{-\epsilon} G_3(\epsilon, x).$$

- **Soft boundary** (if the NNLO strategy is repeated for the other masters):

$$\mathcal{N}_3\Phi_3^{\mathrm{LP}} = \big(y(1-y)\big)^{2-3\epsilon}\bar{z}^{4-6\epsilon} \times \frac{e^{3\epsilon\gamma_E}\Gamma(1-\epsilon)^3}{8\Gamma(3-3\epsilon)^2},$$

$$\frac{e^{3\epsilon\gamma_E}\Gamma(1-\epsilon)^3}{8\Gamma(3-3\epsilon)^2} = \frac{1}{32} + \frac{9\epsilon}{32} + \frac{207-5\pi^2}{128}\epsilon^2 + \frac{972-45\pi^2-68\zeta_3}{128}\epsilon^3 + \frac{488430-31050\pi^2+163\pi^4-73440\zeta_3}{15360}\epsilon^4 + \ldots$$

  For *this* master no boundary is needed, since the exact formula is known. It can also serve as an exact check of the differential-equation solution at any z.

11 Checks

`checks/checks.wl` (Mathematica; output in `checks.log`):

1. the one-dimensional integral (9) against the ${}_2F_1$ form and against (11), for $n = 2, 3, 4$ and five points (ϵ, z, u) with ϵ of both signs: agreement to 10^{-28};
2. $n = 1$: $G_1 = 1$ (symbolic);
3. the identities of the NNLO variables (symbolic);
4. the leading power of w and x;
5. the ϵ and ϵ^2 coefficients of G_2 and G_3: from the series of ${}_2F_1$ and the Euler integral, checked by finite differences; the simplified forms above are identical to the Euler-integral results;
6. the NNLO soft master file against $e^{2\epsilon\gamma_E}\Gamma(1-\epsilon)^2/(4\Gamma(2-2\epsilon)^2)$: identical through ϵ^6;
7. the comment formula of `boundaries.nb`: symbolic identity with $w^{1-\epsilon}/(1+u)[\ldots]^{-\epsilon}$, and the table of Sect. 9;
8. the N^{3}LO leading-power constant;
9. the $p_1 \leftrightarrow p_2$ symmetry $\Phi_n(z, 1/u) = u\,\Phi_n(z, u)$.

`checks/notebook_integrand.wl` (output in `notebook_integrand.log`): the notebook's `INTEGRAND`, with and without $1/Q{\cdot}p_2$, against the light-cone results and F_c (table of (c)). `checks/rest_frame.py` (plain Python; output in `rest_frame.log`): an independent derivation in the partonic rest frame instead of light-cone variables. $P \to Q + K$ two-body, $K \to n$ massless as in Sect. 3, and the angle of Q fixed by the rapidity delta. Compared with (11) in absolute normalisation for $n = 2, 3$, $\epsilon = 0$, 0.3, -0.2 and three (s, z, u) points: agreement to 10^{-9}–10^{-14} (the accuracy of the quadrature).